1) a) EN LA MALLA DE LA DERECHA HAY TRES RESISTENCIAS EN PARALELO, Y POR CADA UNA DE LAS DOS RESISTENCIAS R_1 PASAN 2 A (PUES $R_2 = 2R_1$). LUEGO EN LA RAMA DE LA IZQUIERDA (LA DE LA BATERIA) LA CORRIENTE ES 5 A.

LA RESISTENCIA EQUIVALENTE DEL CIRCUITO ES:

$$R_{ef} = R_1 + \left(\frac{1}{R_1} + \frac{1}{R_1} + \frac{1}{2R_1}\right)^{-1} = \frac{7}{5} R_1 = \frac{15V}{5A} \rightarrow \boxed{R_1 = \frac{15}{7}\,\Omega = 2{,}14\,\Omega}$$

b) $P = VI = 15V \times 5A \rightarrow \boxed{P = 75\,W}$

2) a) POR GAUSS:

$$\vec{E} = \begin{cases} \frac{\rho r}{3\varepsilon_0}\hat{r} & (r < R) \\ \frac{\rho R^3}{3\varepsilon_0 r^2}\hat{r} & (r > R) \end{cases}$$

LUEGO:

$$\frac{\rho}{3\varepsilon_0}\frac{R}{2} = 8\frac{V}{m} \rightarrow \frac{\rho R}{\varepsilon_0} = 48\frac{V}{m}$$

$$|\vec{E}(2R)| = \frac{\rho R^3}{3\varepsilon_0}\frac{1}{4R^2} = \frac{\rho R}{\varepsilon_0}\frac{1}{12} = \boxed{4\frac{V}{m}}$$

b) $$|\Delta V| = \left|\int_{3R}^{2R} \vec{E}\cdot d\vec{l}\right| = \frac{\rho R^3}{3\varepsilon_0}\left(\frac{1}{2R} - \frac{1}{3R}\right) = \frac{\rho R^3}{3\varepsilon_0}\frac{1}{6R} = \frac{\rho R}{\varepsilon_0}\frac{R}{18}$$

$$\rightarrow 1V = 48\frac{V}{m}\frac{R}{18} = \frac{8}{3}R\frac{V}{m} \rightarrow \boxed{R = \frac{3}{8}m = 37{,}5\,cm}$$

c) $$V(4R) - V(3R) = -\int_{3R}^{4R} \vec{E}\cdot d\vec{l} = -\frac{\rho R^3}{3\varepsilon_0}\left(\frac{1}{3R} - \frac{1}{4R}\right) = -\frac{\rho R^2}{36\varepsilon_0}$$

$$W = -q\left[V(4R) - V(3R)\right] = \frac{\rho q R^2}{36\varepsilon_0} = 48\frac{V}{m} \times \frac{3}{8}m \times \frac{5}{36} \times 10^{-6}C = \boxed{2{,}5\,\mu J}$$

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3) a) CAMPO DEL HILO 1 EN X_5:

$$\vec{E}_1(X_5) = k\lambda_1\left[\frac{1}{X_5 - X_2} - \frac{1}{X_5 - X_1}\right]\hat{\imath} =$$

$$= 9\times10^9 \frac{N m^2}{C^2} \times 5\times10^{-6} \frac{C}{m}\left[\frac{1}{0{,}909\,m - 0{,}906\,m} - \frac{1}{0{,}909\,m - 0{,}902\,m}\right]\hat{\imath}$$

$$\vec{E}_1(X_5) = 8{,}57\times10^5 \frac{N}{C}\hat{\imath}$$

CAMPO DEL HILO 2 EN X_5:

$$\vec{E}_2(X_5) = k\lambda_2\left[\frac{1}{X_5 - X_4} - \frac{1}{X_5 - X_3}\right](-\hat{\imath}) =$$

$$= 9\times10^9 \frac{N m^2}{C^2} \times (-3)\times10^{-6} \frac{C}{m}\left[\frac{1}{0{,}909\,m - 0{,}919\,m} - \frac{1}{0{,}909\,m - 0{,}913\,m}\right](-\hat{\imath})$$

$$\vec{E}_2(X_5) = 4{,}05\times10^5 \frac{N}{C}\hat{\imath}$$

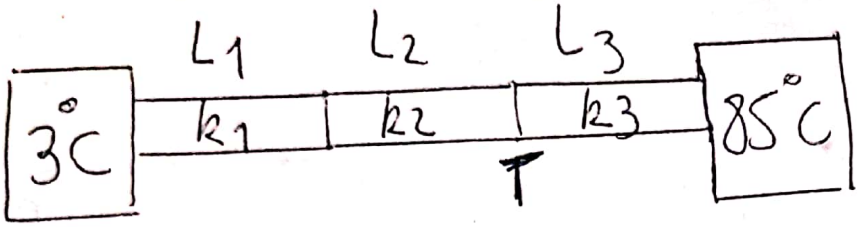
$$\vec{E}(X_5) = \vec{E}_1(X_5) + \vec{E}_2(X_5) = \boxed{12{,}62\times10^5 \frac{N}{C}\hat{\imath}}$$

b) $\vec{F}_q = q\vec{E}(X_5) = -6\times10^{-6}\,C \times 12{,}62\times10^5 \frac{N}{C}\hat{\imath}$

$$\boxed{\vec{F}_q = -7{,}57\,N\,\hat{\imath}}$$

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4) a)



$A = \pi \times (0{,}014\,m)^2 = 0{,}0006158\,m^2$

$R_1 = \frac{L_1}{k_1 A} = \frac{0{,}33\,m}{320\,\frac{W}{m°C} \times 0{,}0006158\,m^2} = 1{,}6748\,\frac{°C}{W}$

$R_2 = \frac{L_2}{k_2 A} = 1{,}6666\,\frac{°C}{W}$ $\quad R_3 = \frac{L_3}{k_3 A} = 1{,}6239\,\frac{°C}{W}$

$R_{ef} = R_1 + R_2 + R_3 = 4{,}9653\,\frac{°C}{W} = \frac{85°C - 3°C}{I} \rightarrow I = 16{,}5146\,W$

$85 - T = I R_3 \rightarrow T = 85°C - 16{,}5146\,W \times 1{,}6239\,\frac{°C}{W} = \boxed{58{,}18°C}$

b) $\tilde{I} = 0{,}9\,I \rightarrow \frac{\Delta T}{\tilde{R}} = 0{,}9\,\frac{\Delta T}{R_{ef}} \rightarrow \tilde{R} = \frac{R_{ef}}{0{,}9} = 5{,}517\,\frac{°C}{W}$

$\rightarrow R_1 + R_2 + \tilde{R}_3 = 5{,}517\,\frac{°C}{W} \rightarrow 3{,}3414\,\frac{°C}{W} + \tilde{R}_3 = 5{,}517\,\frac{°C}{W}$

$\rightarrow \tilde{R}_3 = 2{,}1756\,\frac{°C}{W} = \frac{L_3}{\tilde{k}_3 A} \rightarrow \tilde{k}_3 = \frac{0{,}51\,m}{2{,}1756\,\frac{°C}{W} \times 0{,}0006158\,m^2}$

$\rightarrow \boxed{\tilde{k}_3 = 380{,}67\,\frac{W}{m°C}}$

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